



We are on a mission to address the digital skills gap for 10 Million+ young professionals, train and empower them to forge a career path into future tech

Selenium Setup Procedure

APRIL, 2026



Contents

- Install Python.
- Install Visual Studio Code IDE.
- Setting up a Virtual Environment
- Install Python Extension

Install Python

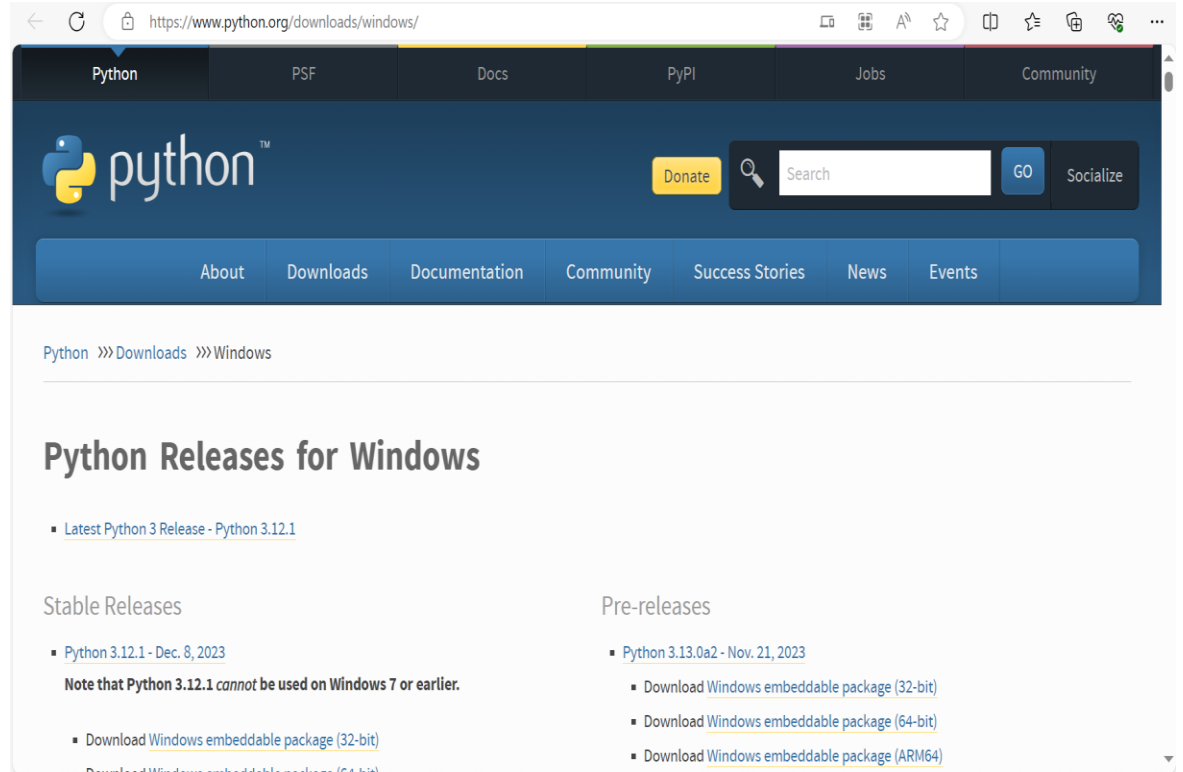


Install Python

Installing Python and Setting up of Environment Variables

- Install Python on your computer
- Download and install the Python by using

<https://www.python.org/downloads/windows/>



Install Python

Installing Python and Setting up of Environment Variables

Files

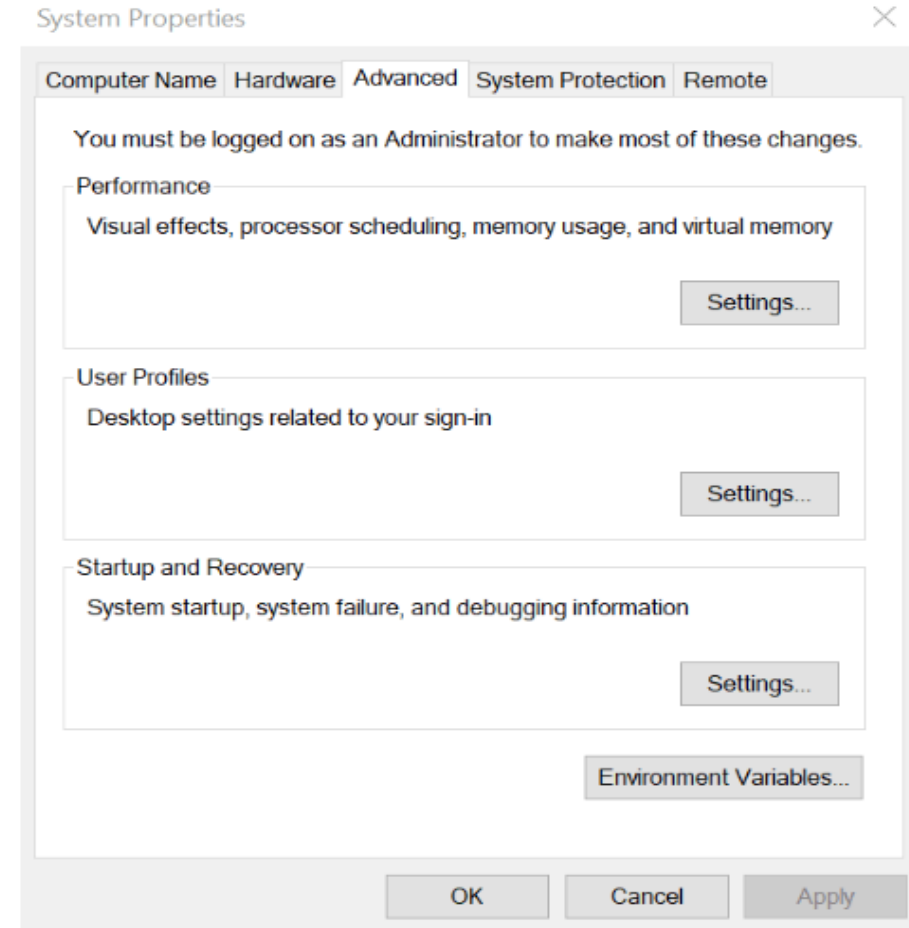
Version	Operating System	Description	MD5 Sum	File Size	GPG	Sigstore
Gzipped source tarball	Source release		51c5c22dcabc698483734dff5c8028606	27053241	SIG	.sigstore
XZ compressed source tarball	Source release		50f827c800483776c8ef86e6a53831fa	20583448	SIG	.sigstore
macOS 64-bit universal2 installer	macOS	for macOS 10.9 and later	eae2d617cbd978a4a6c167924b287572	45377529	SIG	.sigstore
Windows embeddable package (32-bit)	Windows		acc28815c74facc402469e917c8f8433	9849182	SIG	.sigstore
Windows embeddable package (64-bit)	Windows		019788d34af2c60a7be45bf8273e361f	11061655	SIG	.sigstore
Windows embeddable package (ARM64)	Windows		18058aa3c8ccbf3e8fee53386d38c711	10277136	SIG	.sigstore
Windows installer (32 -bit)	Windows		37a89a09138888e6331ec279d68fea8e	25253448	SIG	.sigstore
Windows installer (64-bit)	Windows	Recommended	3e3b6550e58772d324f7519bfa8066dc	26589696	SIG	.sigstore
Windows installer (ARM64)	Windows	Experimental	25fb741f175dc98d5630520f2df931ec	25806784	SIG	.sigstore

Install Python

Installing Python and Setting up of Environment Variables

- Once the installation is complete, set the environment variables. The steps are as below:

- 1.Right-click on Computer.
- 2.Click on 'Properties'.
- 3.Click on 'Advanced system settings'.
- 4.On System Properties, select the 'Advanced' Tab.

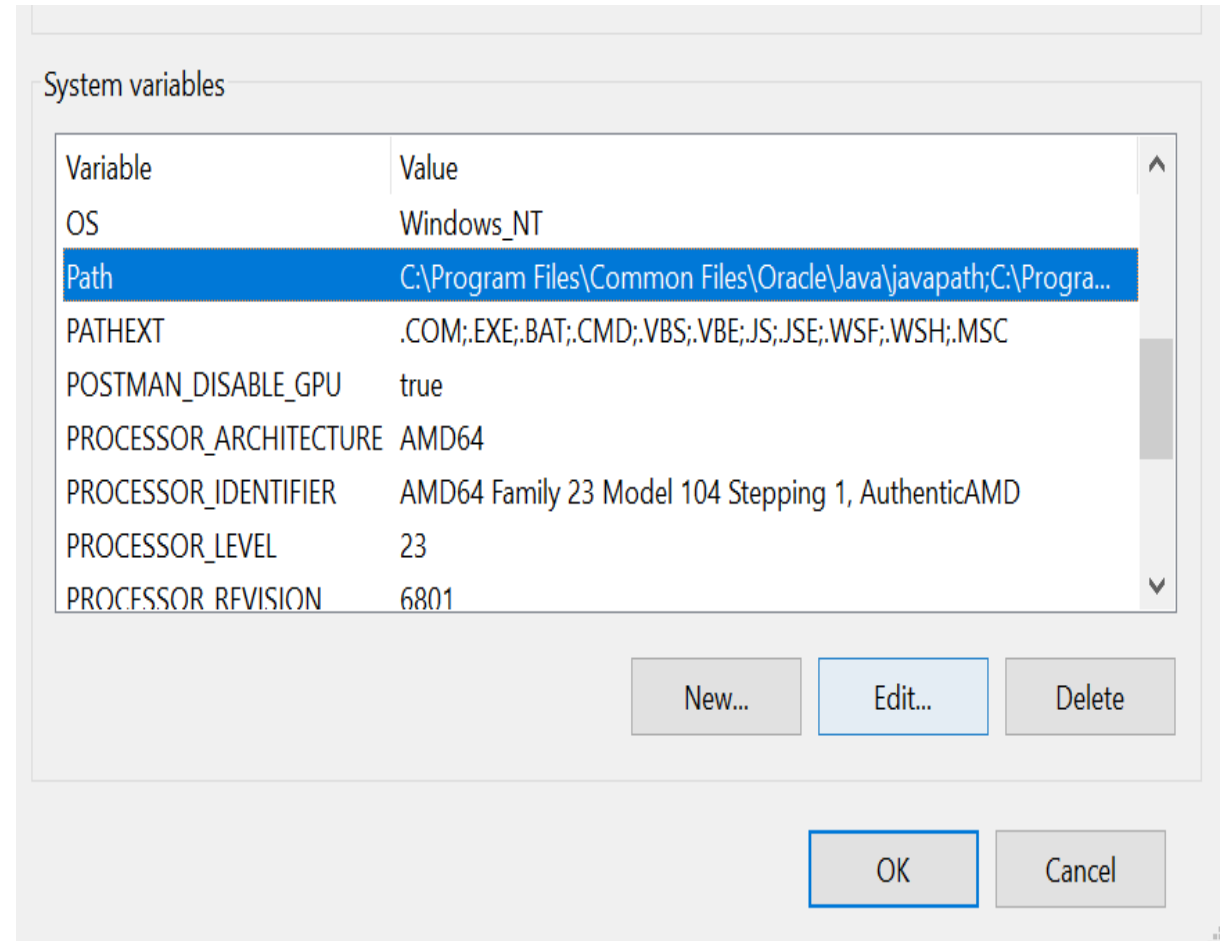


Install Python

Installing Python and Setting up of Environment Variables

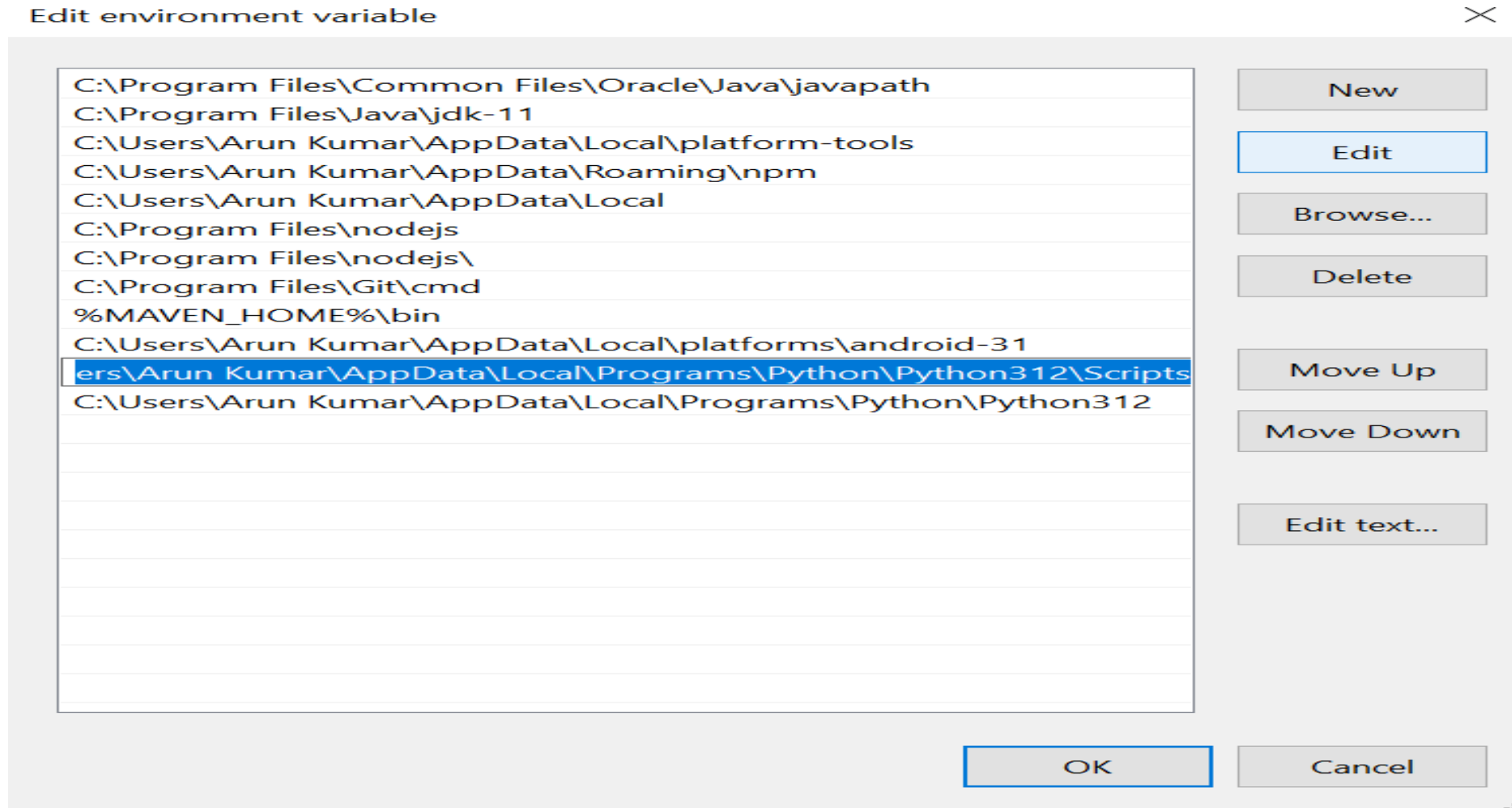
5. Click on 'Environment Variables.'
6. Copy the path where the Python is installed.
(E.g.
C:\Users\ArunKumar\AppData\Local\Programs\Python\Python312\Scripts)

(**C:\Users\ArunKumar\AppData\Local\Programs\Python\Python312**)
7. In the opened Environment Variables window,
select 'New' under 'System variables'.
8. Set the path. If 'Path' is already existing, then edit it.



Install Python

Installing Python and Setting up of Environment Variables



Install Python

Installing Python and Setting up of Environment Variables

- Open the command prompt and run the following command **Python --version** to verify that Python has been successfully installed in your system.



```
Administrator: Command Prompt
C:\Windows\system32>Python --version
Python 3.12.1
C:\Windows\system32>
```

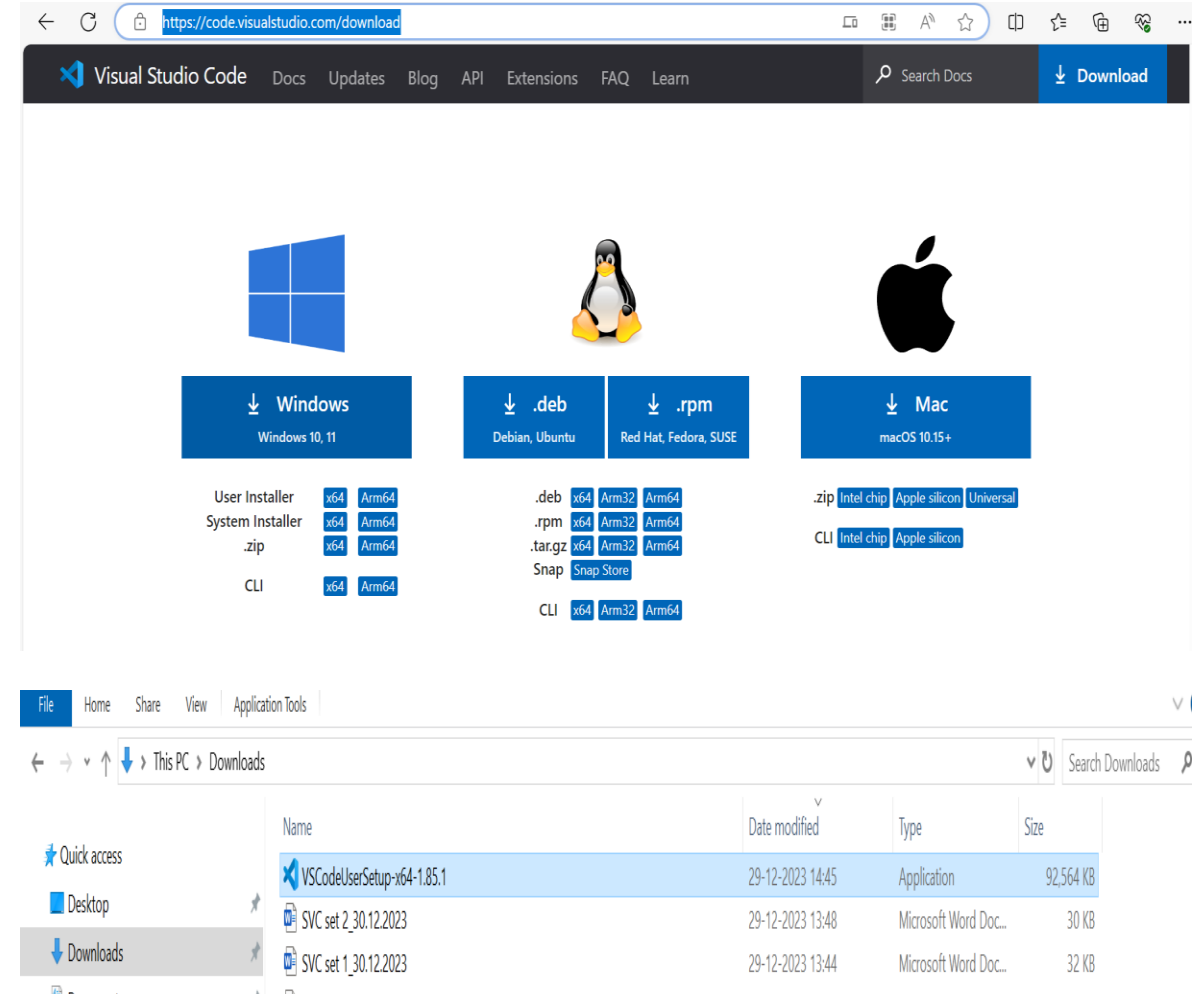
Installing Visual Studio Code



Installing Visual Studio Code

Installing VS code

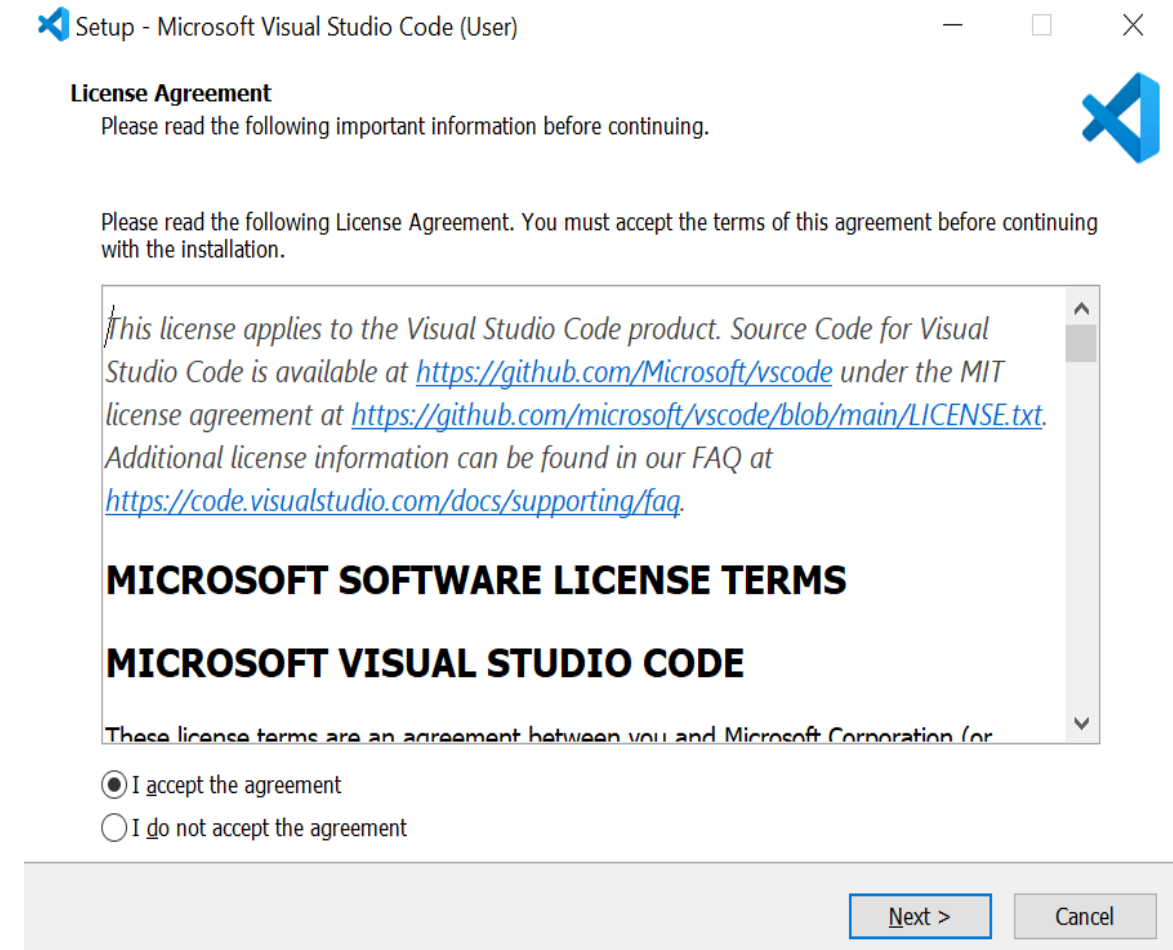
- Download latest version of "VS code" by using <https://code.visualstudio.com/download>
- Be sure to choose correctly between Windows 32 Bit and 64 Bit versions.
- You should be able to download an exe file named "**VSCodeUserSetup-x64-1.85.1**" for Setup.



Installing Visual Studio Code

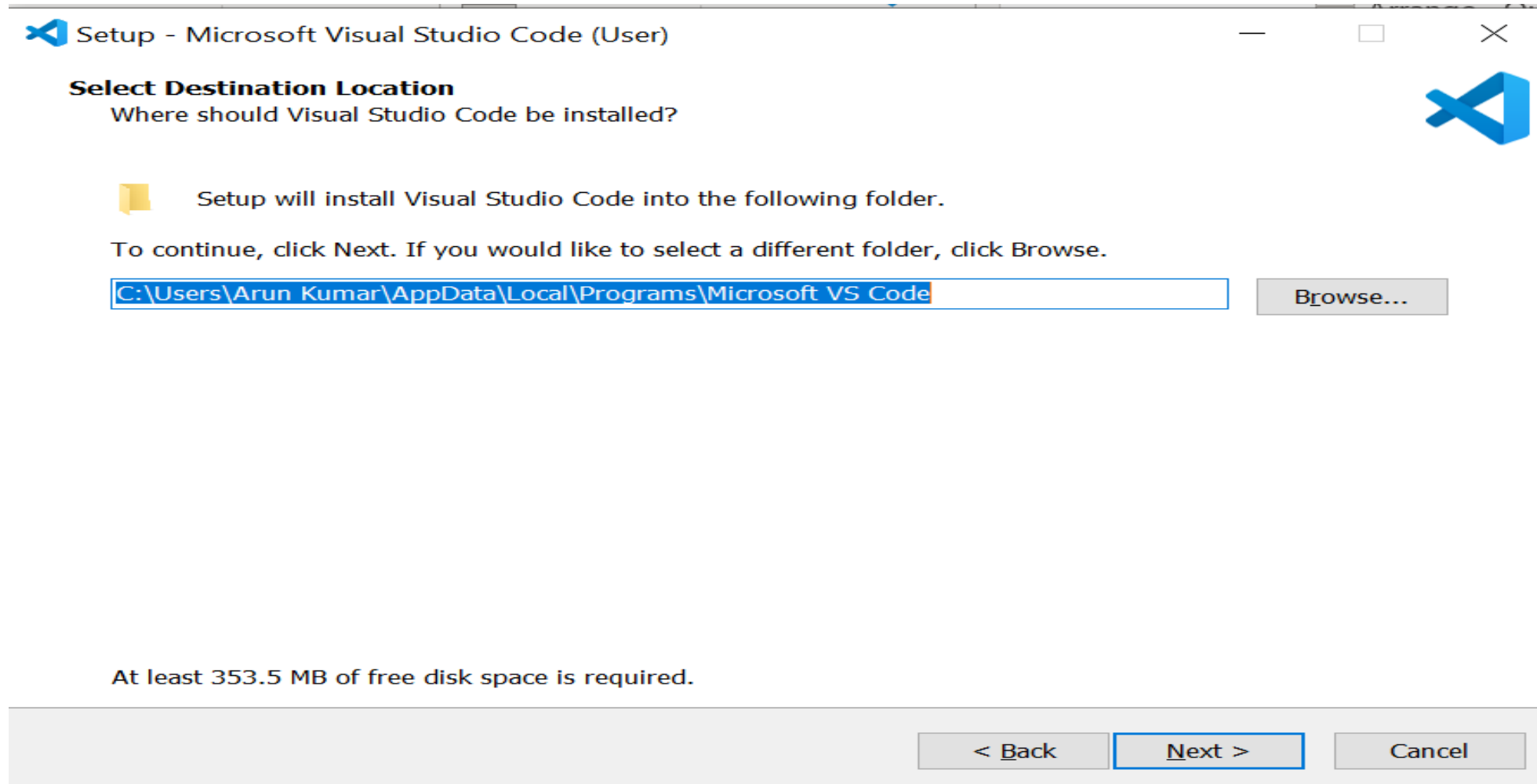
Installing VS code (contd.)

- Double-click on file to Install the VS Code.
- License Agreement window will open. **Click I accept the agreement and NEXT.**
- After that, a new window will open which will ask for the location to install VS Code.



Installing Visual Studio Code

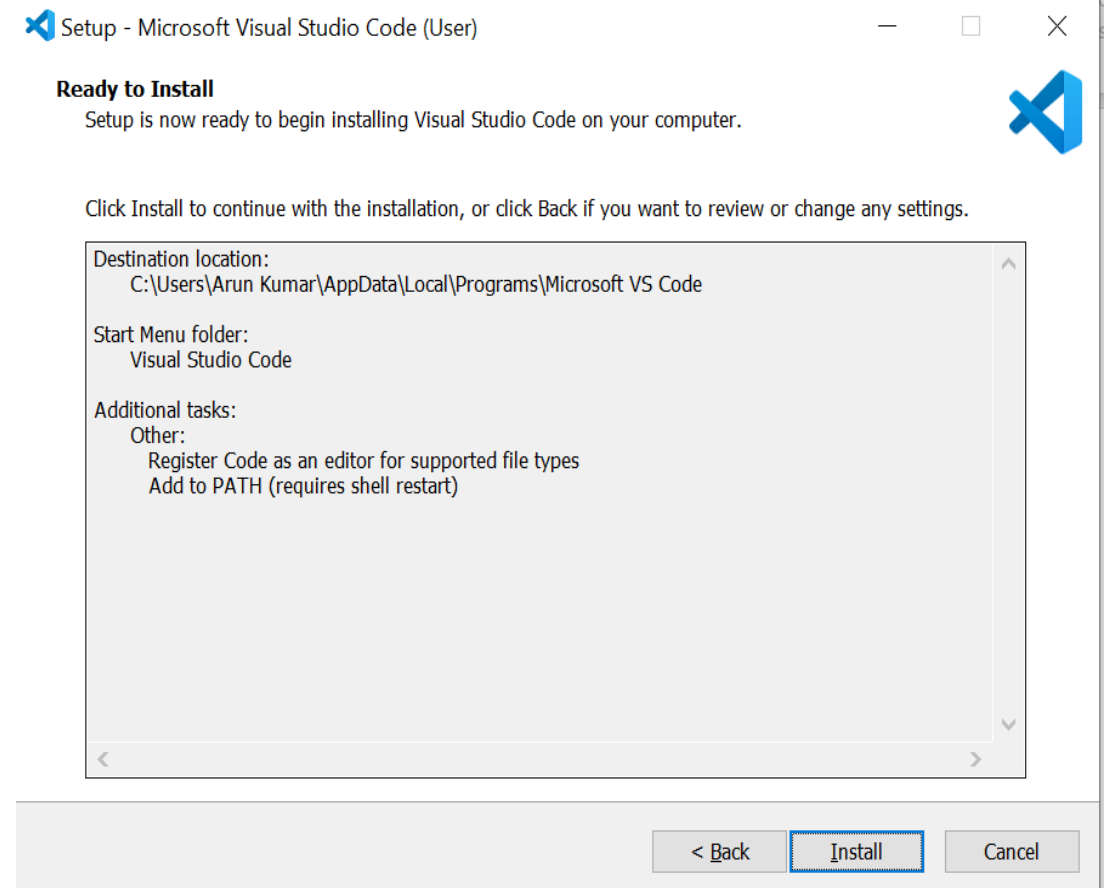
Installing VS code (contd.)



Installing Visual Studio Code

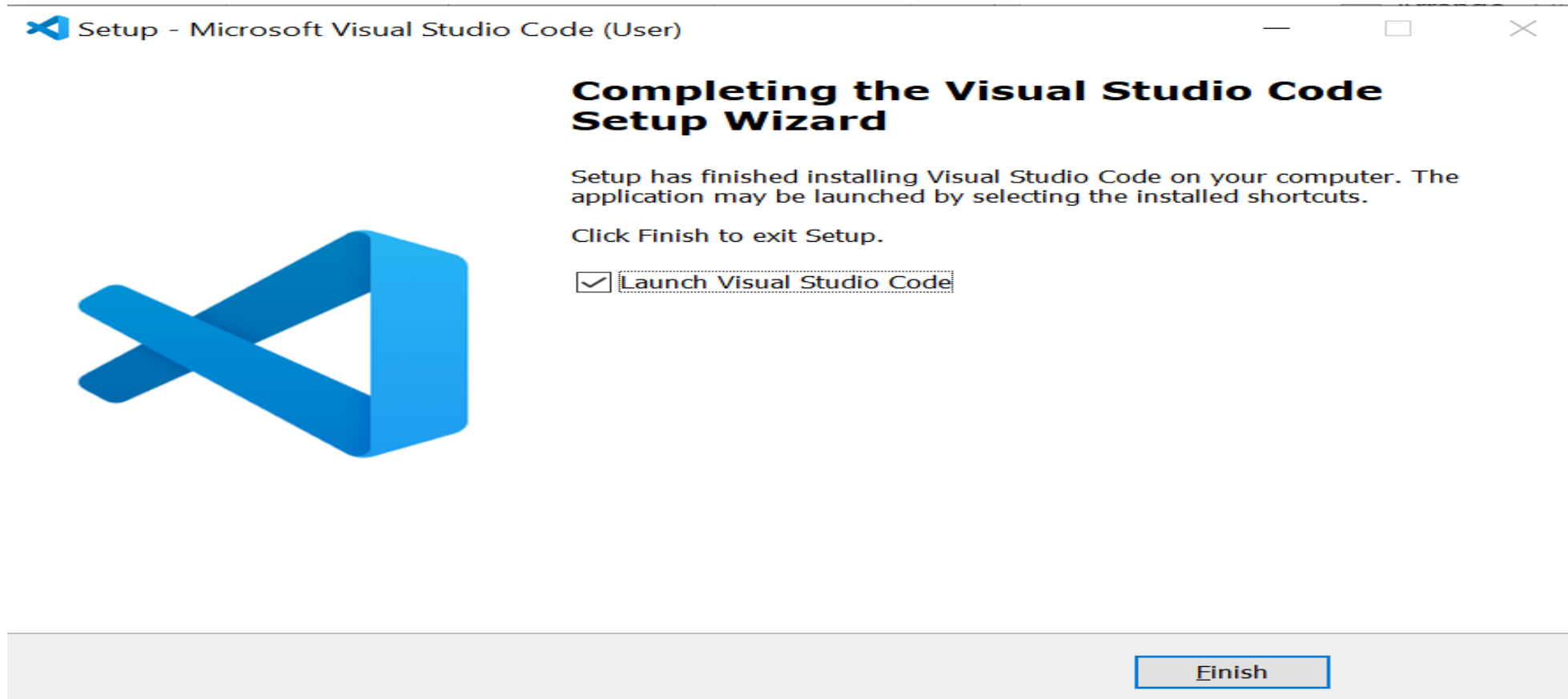
Installing VS code (contd.)

- After that **Select Start Menu Folder** window will appear. **Click Next.**
- After that **Select Additional Tasks** window will appear, go with the default selected options and **Click Next.**
- **Click Install in the Ready to Install Window**



Installing Visual Studio Code

Installing VS code (contd.)



Setting up a Virtual Environment



Setting up a Virtual Environment

Virtual Environment

- When developing software with Python, a basic approach is to install Python on your machine, install all your required libraries via the terminal, write all your code in a single.py file or notebook, and run your Python program in the terminal.
- This works fine for simple Python scripting projects.
- But in complex software development projects, like building a Python library, an API, or software development kit, often you will be working with multiple files, multiple packages, and dependencies.
- As a result, you will need to isolate your Python development environment for that particular project.

Setting up a Virtual Environment

Virtual Environment

- A **virtual environment** is a Python environment such that the Python interpreter, libraries and scripts installed into it are isolated from those installed in other virtual environments, and (by default) any libraries installed in a “system” Python, i.e., one which is installed as part of your operating system.

Setting up a Virtual Environment

How to Install a Virtual Environment using Venv

- **Virtualenv** is a tool to set up your Python environments. Since Python 3.3, a subset of it has been integrated into the standard library under the venv module. You can install venv to your host Python by running this command in your terminal:

pip install virtualenv

- To use venv in your project, in your terminal, create a new project folder, cd to the project folder in your terminal, and run the following command:

```
python<version> -m venv <virtual-environment-name>
```

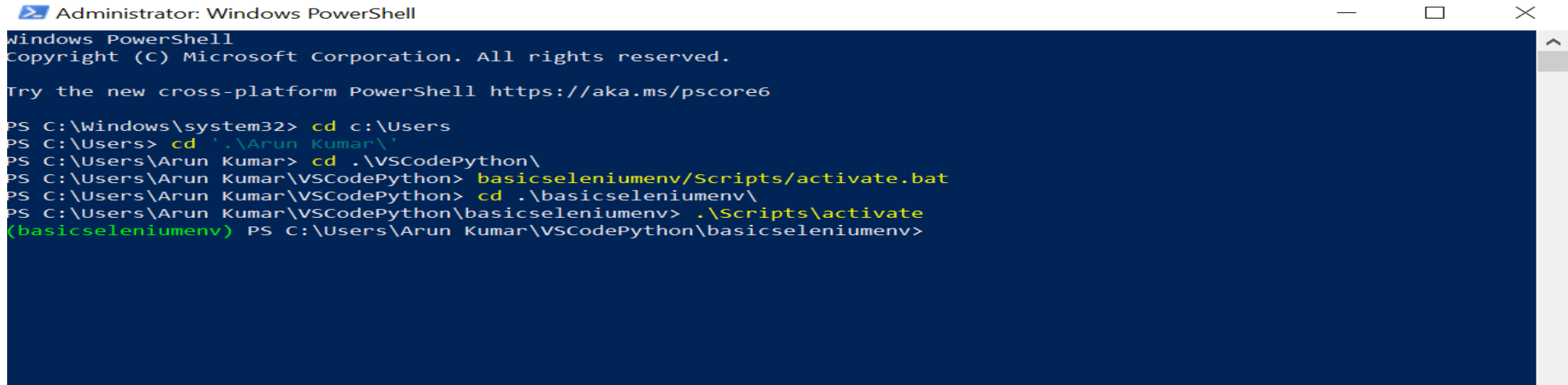
- Eg : python3.12.1 -m venv basicseleniumenv

Setting up a Virtual Environment

How to activate the Virtual Environment

- To activate your virtual environment on Windows, you will need to run the following code below.
- Go to the location of virtual environment in Power Shell using admin rights and type

`.\Scripts\activate`



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

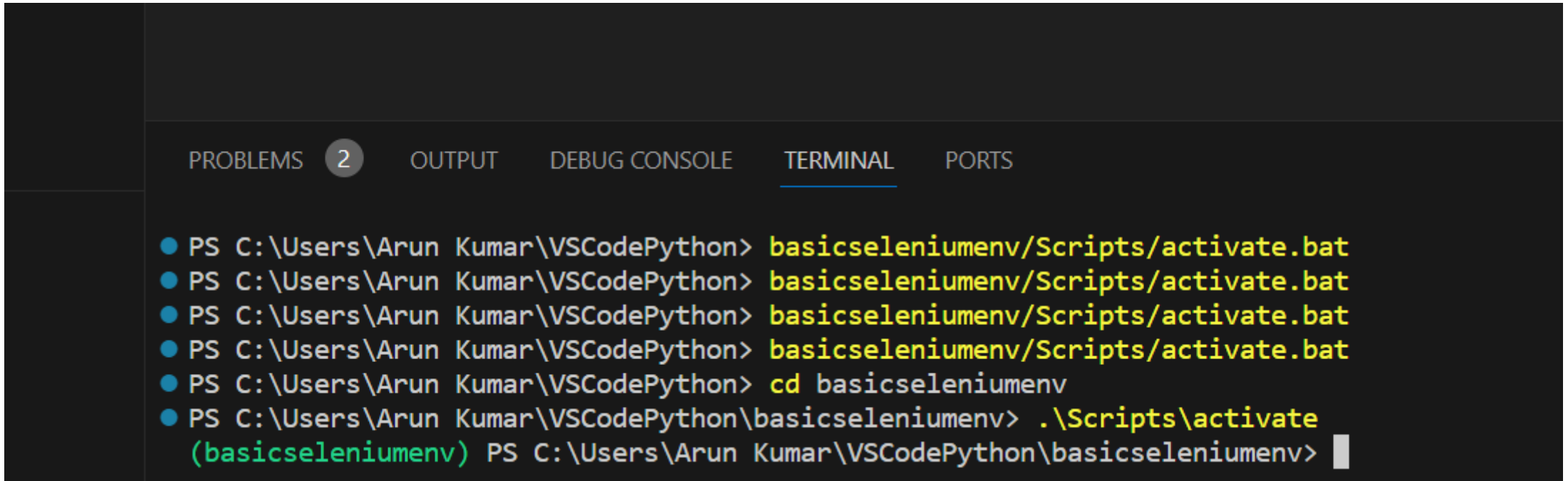
Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> cd c:\Users
PS C:\Users> cd '.\Arun Kumar\'
PS C:\Users\Arun Kumar> cd .\VSCodePython\
PS C:\Users\Arun Kumar\VSCodePython> basicseleniumenv/Scripts/activate.bat
PS C:\Users\Arun Kumar\VSCodePython> cd .\basicseleniumenv\
PS C:\Users\Arun Kumar\VSCodePython\basicseleniumenv> .\Scripts\activate
(basicseleniumenv) PS C:\Users\Arun Kumar\VSCodePython\basicseleniumenv>
```


Setting up a Virtual Environment

How to activate the Virtual Environment

Repeat the same in the VS code terminal



The screenshot shows the VS Code interface with the 'TERMINAL' tab selected. The terminal displays a series of commands to set up and activate a virtual environment named 'basic seleniumenv'.

```
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

● PS C:\Users\Arun Kumar\VSCodePython> basic seleniumenv/Scripts/activate.bat
● PS C:\Users\Arun Kumar\VSCodePython> basic seleniumenv/Scripts/activate.bat
● PS C:\Users\Arun Kumar\VSCodePython> basic seleniumenv/Scripts/activate.bat
● PS C:\Users\Arun Kumar\VSCodePython> basic seleniumenv/Scripts/activate.bat
● PS C:\Users\Arun Kumar\VSCodePython> cd basic seleniumenv
● PS C:\Users\Arun Kumar\VSCodePython\basic seleniumenv> .\Scripts\activate
(basic seleniumenv) PS C:\Users\Arun Kumar\VSCodePython\basic seleniumenv> |
```

Setting up a Virtual Environment

How to activate the Virtual Environment

Is the virtual environment working?

- We have activated our virtual environment, now how do we confirm that our project is in fact isolated from our host Python.
- First we check the list of packages installed in our virtual environment by running the code below in the activated virtual environment.

pip list

- You will notice only two packages – **pip** and **setuptools**, which are the base packages that come default with a new virtual environment

Setting up a Virtual Environment

How to Install Libraries in a Virtual Environment

- To install new libraries, you can easily just pip install the libraries. The virtual environment will make use of its own pip, so you don't need to use pip3.

OR

- you can generate a text file listing all your project dependencies by running the code below:

```
pip freeze > requirements.txt
```

Why is a requirements file important to your project?

- Instead of having to install each dependency one by one, they could just run the code below to install all your dependencies within their own copy of the project:

```
~ pip install -r requirements.txt
```

Installing Selenium WebDriver

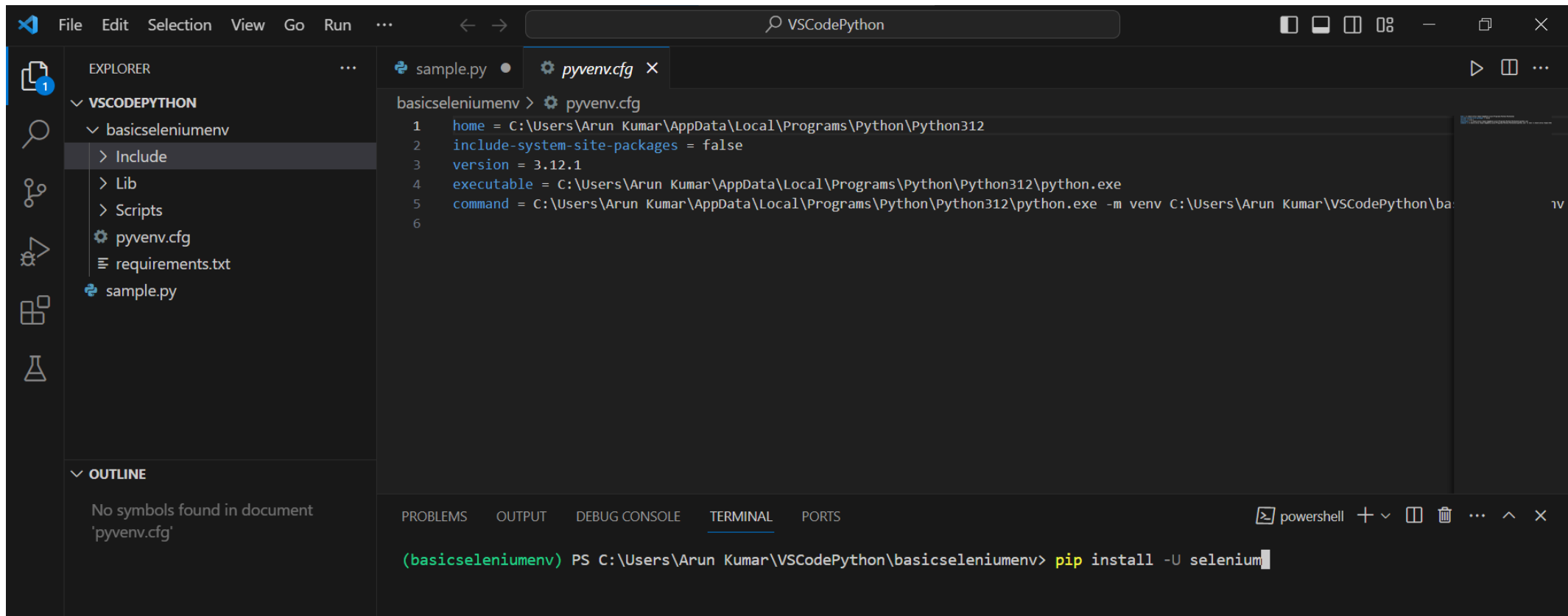


Selenium Webdriver

Installing Selenium webdriver

- Go to terminal in VS Code and type the following command :

pip install -U selenium



The screenshot shows the VS Code interface with the following components:

- EXPLORER:** Shows the project structure for 'VSCODEPYTHON'. The 'basic seleniumenv' folder is expanded, showing subfolders 'Include', 'Lib', and 'Scripts', and files 'pyenv.cfg', 'requirements.txt', and 'sample.py'.
- pyenv.cfg:** The file is open in the editor, showing the following configuration:


```

1  home = C:\Users\Arun Kumar\AppData\Local\Programs\Python\Python312
2  include-system-site-packages = false
3  version = 3.12.1
4  executable = C:\Users\Arun Kumar\AppData\Local\Programs\Python\Python312\python.exe
5  command = C:\Users\Arun Kumar\AppData\Local\Programs\Python\Python312\python.exe -m venv C:\Users\Arun Kumar\VSCodePython\ba
6

```
- TERMINAL:** The terminal is open at the bottom, showing the command being executed:


```

(basic seleniumenv) PS C:\Users\Arun Kumar\VSCodePython\basic seleniumenv> pip install -U selenium

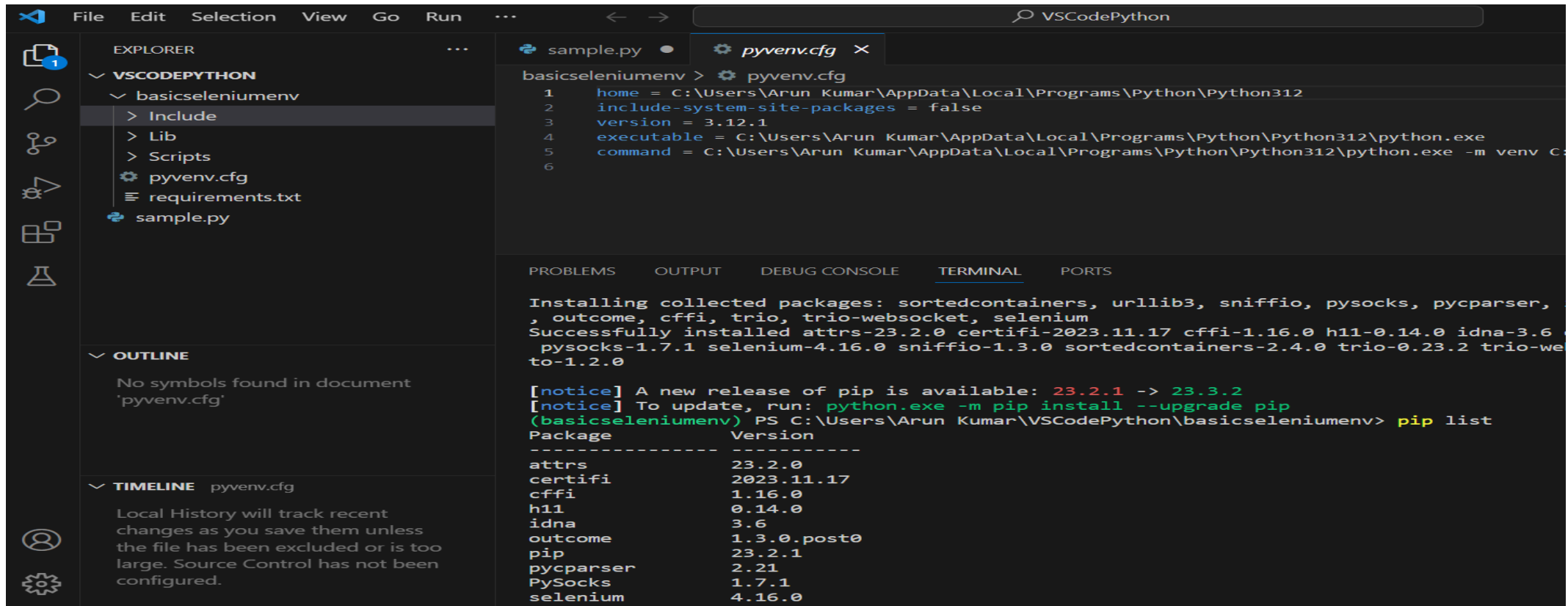
```

Selenium Webdriver

Installing Selenium webdriver

- Type the following command :

pip list



```

basic seleniumenv > pyvenv.cfg
1  home = C:\Users\Arun Kumar\AppData\Local\Programs\Python\Python312
2  include-system-site-packages = false
3  version = 3.12.1
4  executable = C:\Users\Arun Kumar\AppData\Local\Programs\Python\Python312\python.exe
5  command = C:\Users\Arun Kumar\AppData\Local\Programs\Python\Python312\python.exe -m venv C:
6

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

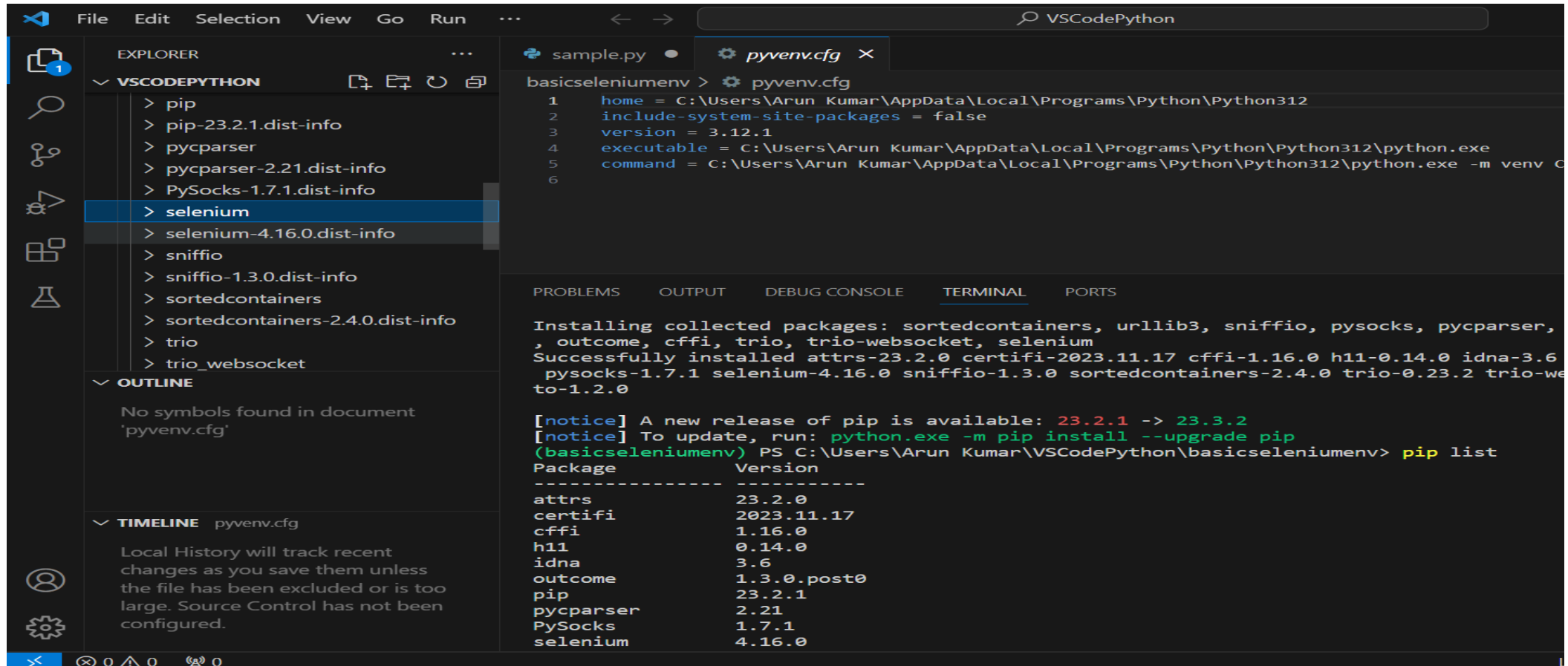
Installing collected packages: sortedcontainers, urllib3, sniffio, pysocks, pycparser,
, outcome, cffi, trio, trio-websocket, selenium
Successfully installed attrs-23.2.0 certifi-2023.11.17 cffi-1.16.0 h11-0.14.0 idna-3.6
pysocks-1.7.1 selenium-4.16.0 sniffio-1.3.0 sortedcontainers-2.4.0 trio-0.23.2 trio-we
to-1.2.0

[notice] A new release of pip is available: 23.2.1 -> 23.3.2
[notice] To update, run: python.exe -m pip install --upgrade pip
(basic seleniumenv) PS C:\Users\Arun Kumar\VSCodePython\basicseleniumenv> pip list
Package            Version
-----
attrs              23.2.0
certifi            2023.11.17
cffi               1.16.0
h11               0.14.0
idna               3.6
outcome           1.3.0.post0
pip               23.2.1
pycparser         2.21
PySocks           1.7.1
selenium          4.16.0
  
```


Selenium Webdriver

Installing Selenium webdriver

- Check whether selenium is installed in your project:



The screenshot shows the VS Code interface with the following components:

- EXPLORER:** A list of installed packages in the `VSODEPYTHON` environment. `selenium` is highlighted.
- pyvenv.cfg:** The configuration file for the virtual environment, showing the Python path and command.
- TERMINAL:** The output of the `pip list` command, showing the installed packages and their versions.

```

basic seleniumenv > pyvenv.cfg
1  home = C:\Users\Arun Kumar\AppData\Local\Programs\Python\Python312
2  include-system-site-packages = false
3  version = 3.12.1
4  executable = C:\Users\Arun Kumar\AppData\Local\Programs\Python\Python312\python.exe
5  command = C:\Users\Arun Kumar\AppData\Local\Programs\Python\Python312\python.exe -m venv C
6

Installing collected packages: sortedcontainers, urllib3, sniffio, pysocks, pycparser,
, outcome, cffi, trio, trio-websocket, selenium
Successfully installed attrs-23.2.0 certifi-2023.11.17 cffi-1.16.0 h11-0.14.0 idna-3.6
pysocks-1.7.1 selenium-4.16.0 sniffio-1.3.0 sortedcontainers-2.4.0 trio-0.23.2 trio-we
to-1.2.0

[notice] A new release of pip is available: 23.2.1 -> 23.3.2
[notice] To update, run: python.exe -m pip install --upgrade pip
(basic seleniumenv) PS C:\Users\Arun Kumar\VSCodePython\basic seleniumenv> pip list
Package          Version
-----
attrs            23.2.0
certifi          2023.11.17
cffi             1.16.0
h11              0.14.0
idna             3.6
outcome          1.3.0.post0
pip              23.2.1
pycparser        2.21
PySocks          1.7.1
selenium         4.16.0
  
```

Installing Python Extension



Install Python Extension

Python Extension

- Install the following Python extension:
1. Python v2023.22.1(Microsoft)
 2. Code Runner v0.12.1
 3. Pylance v2023.12.1
 4. Auto Import v1.5.4



Selenium Project Creation

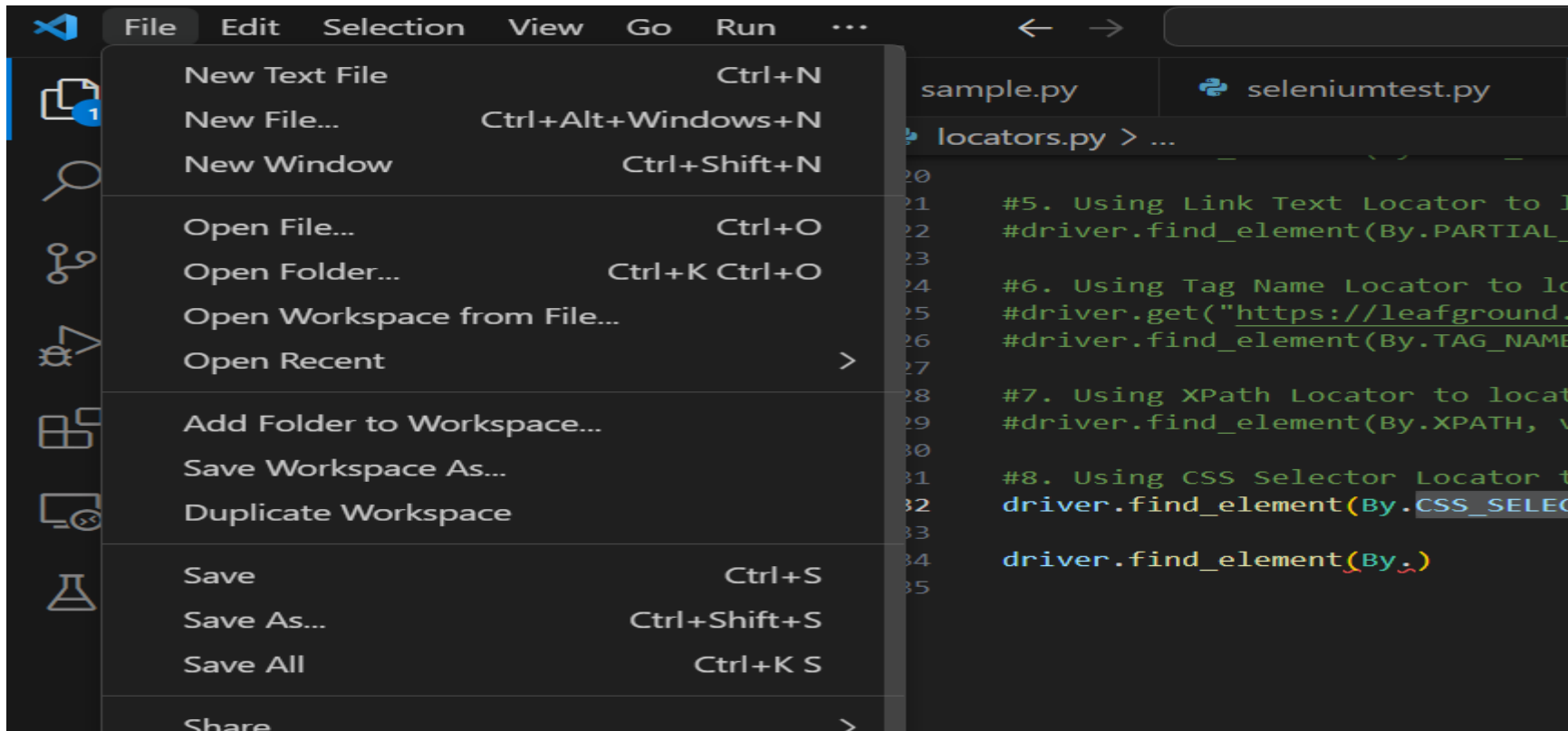


Selenium Project Creation

Selenium Project Creation

Step 1: Create a folder with your project name (like workspace where all the project related files will be stored)

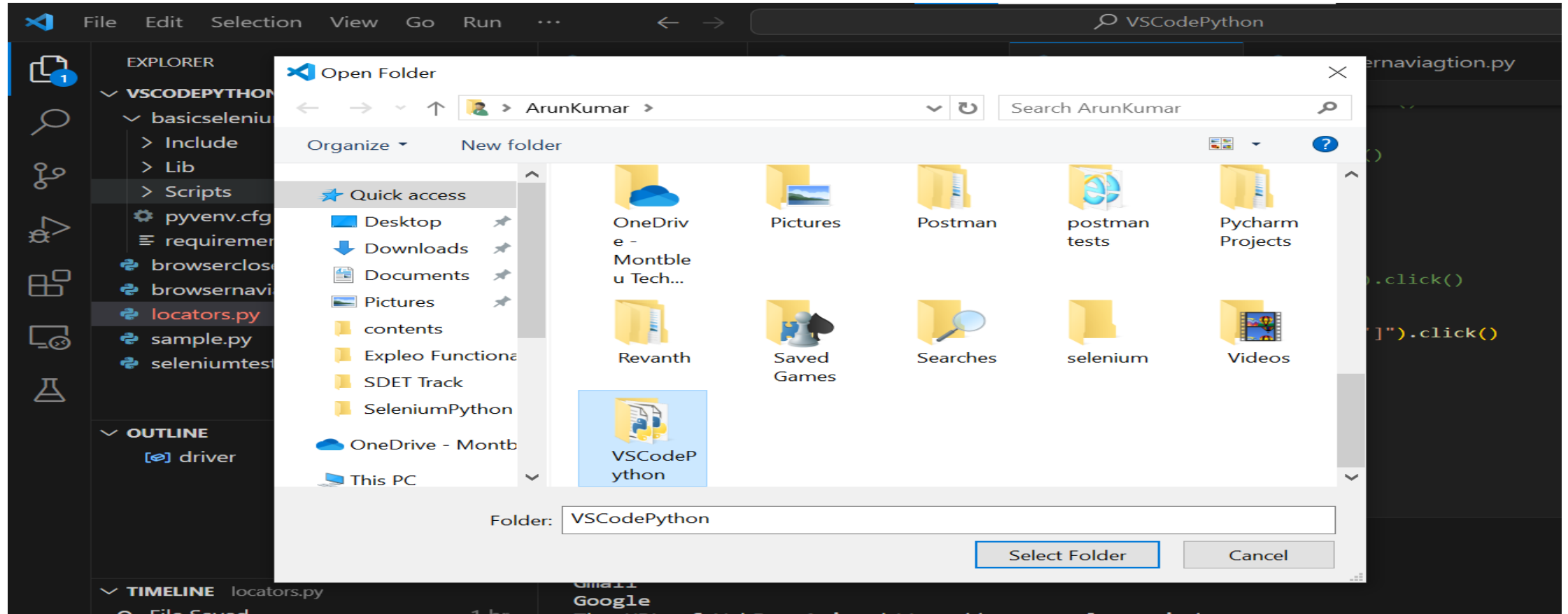
Go to File > Open Folder.



Selenium Project Creation

Selenium Project Creation

Step 2: Choose the folder created for the project.

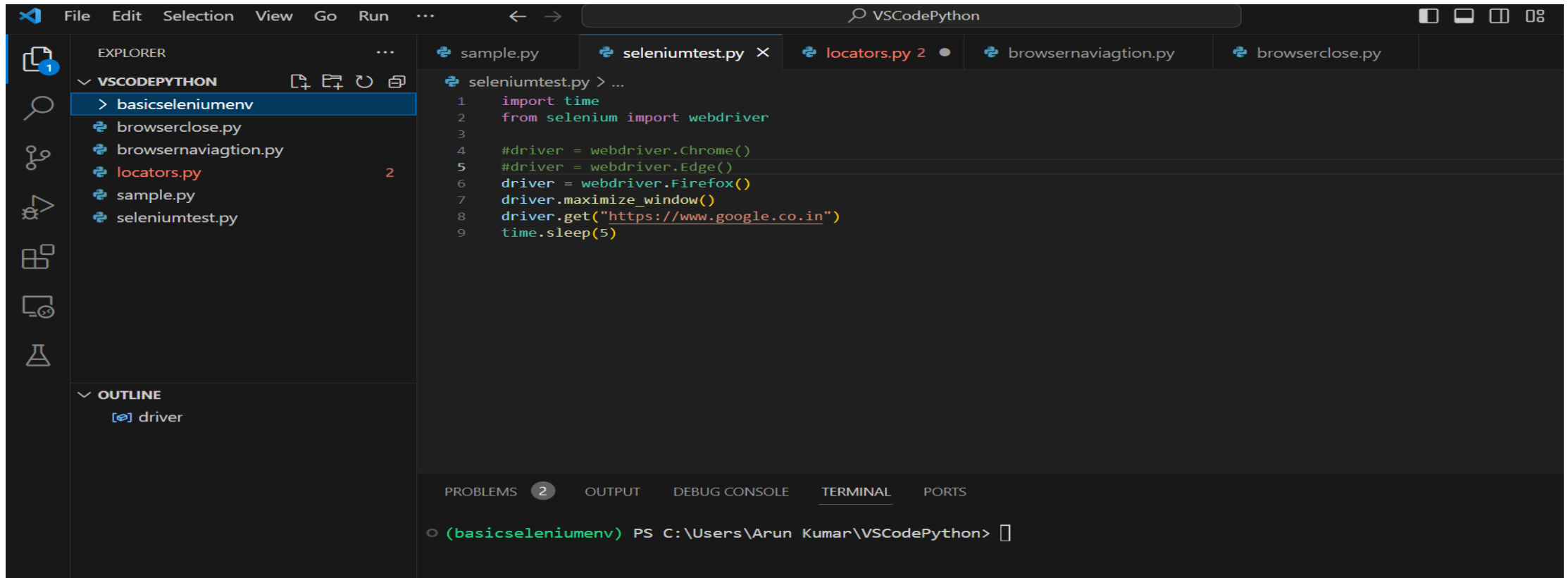


Selenium Project Creation

Selenium Project Creation

Step 3: Create the virtual environment using pipenv

Step 3.1 : Activate the virtual environment



The screenshot displays the Visual Studio Code (VS Code) interface for a Selenium project. The Explorer sidebar on the left shows the 'VSCODEPYTHON' workspace with the following files: `basicseleniumenv` (selected), `browserclose.py`, `browsernavigtion.py`, `locators.py`, `sample.py`, and `seleniumtest.py`. The Outline sidebar below the Explorer shows a 'driver' object. The main editor area displays the `seleniumtest.py` file with the following Python code:

```
1 import time
2 from selenium import webdriver
3
4 #driver = webdriver.Chrome()
5 #driver = webdriver.Edge()
6 driver = webdriver.Firefox()
7 driver.maximize_window()
8 driver.get("https://www.google.co.in")
9 time.sleep(5)
```

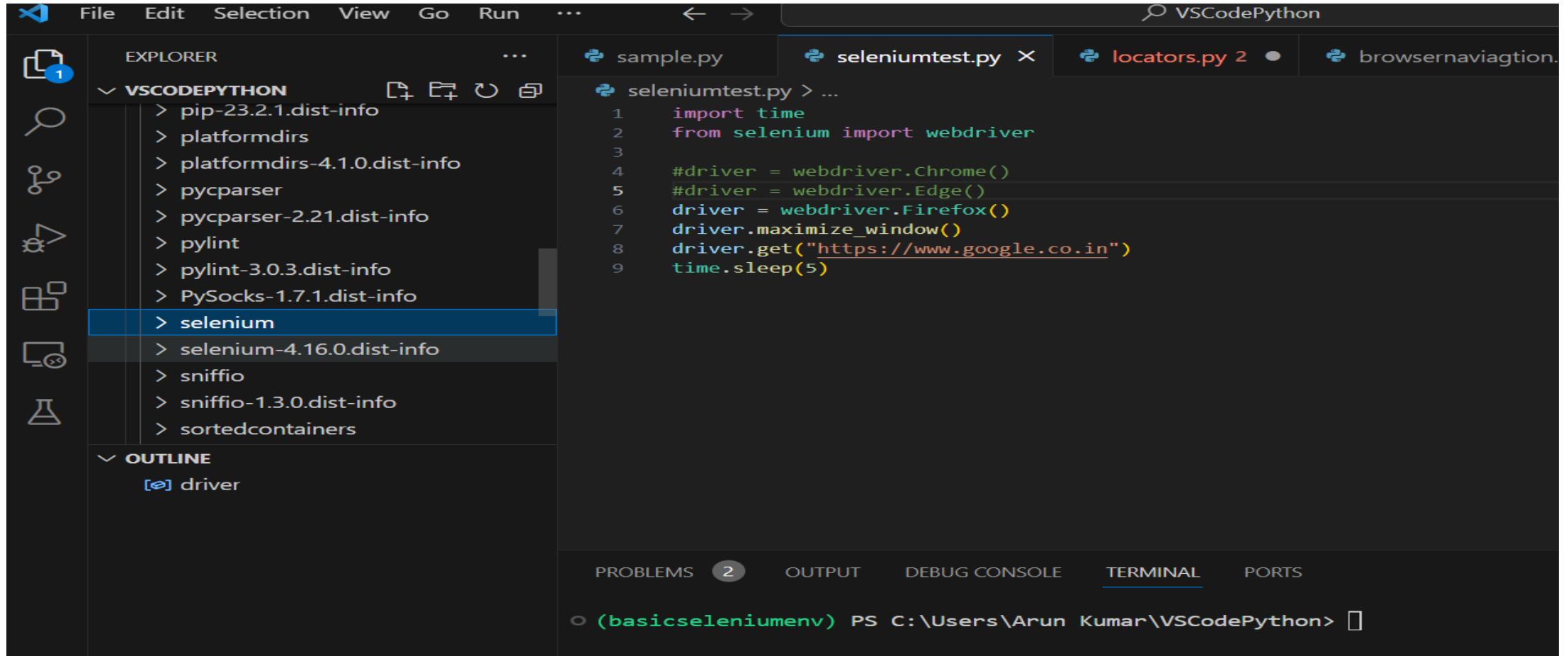
The terminal at the bottom shows the command prompt for the 'basicseleniumenv' environment:

```
(basicseleniumenv) PS C:\Users\Arun Kumar\VSCodePython>
```

Selenium Project Creation

Selenium Project Creation

- **Step 4: Install all the required libraries and packages in the virtual environment**



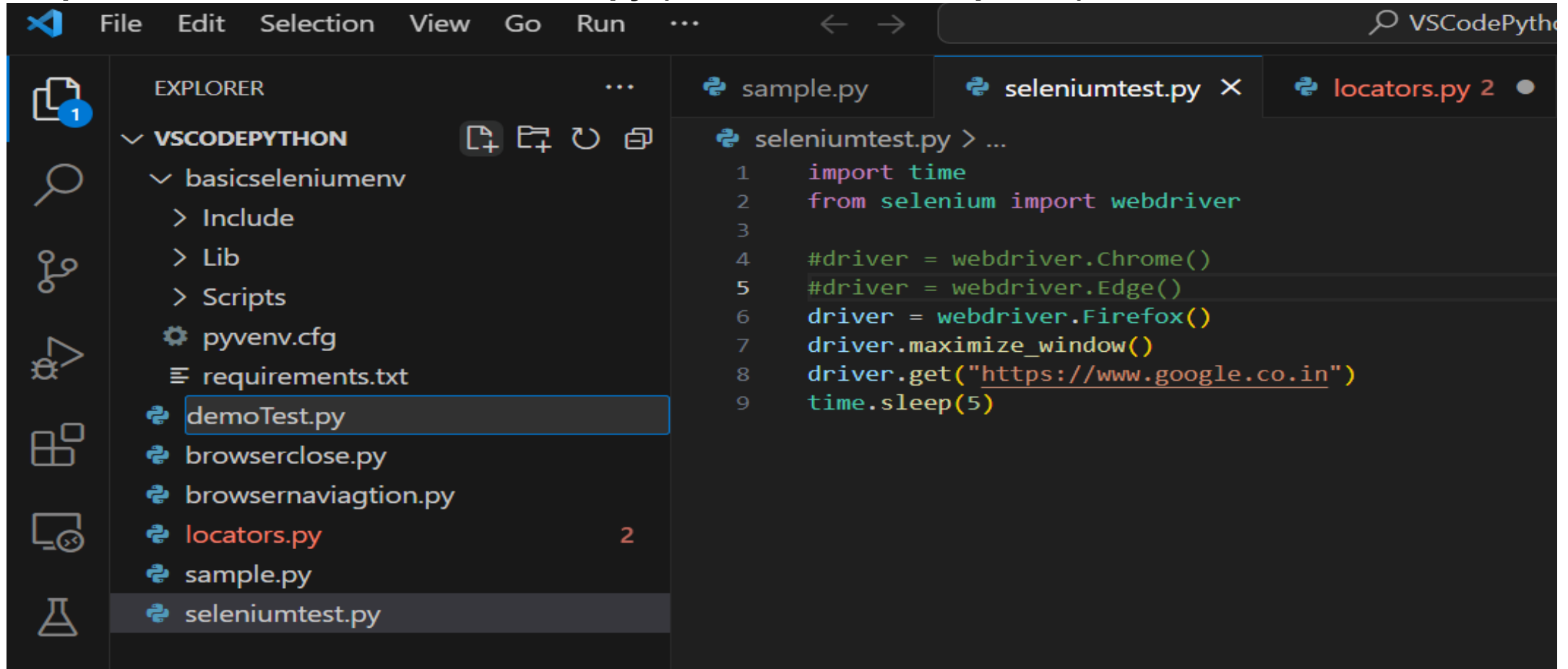
The screenshot displays the Visual Studio Code (VS Code) interface during the Selenium project setup. The Explorer panel on the left shows the 'VSCODEPYTHON' environment with a list of installed packages. The 'selenium' package is highlighted. The Editor panel shows the 'seleniumtest.py' file with the following Python code:

```
1 import time
2 from selenium import webdriver
3
4 #driver = webdriver.Chrome()
5 #driver = webdriver.Edge()
6 driver = webdriver.Firefox()
7 driver.maximize_window()
8 driver.get("https://www.google.co.in")
9 time.sleep(5)
```

The Terminal panel at the bottom shows the command prompt for the 'basic selenium env' environment, with the current directory set to 'C:\Users\Arun Kumar\VSCodePython'.

Selenium Test Script File Creation

Step 5: Create a file with extension .py (Selenium Test script File)



THANK YOU